

Chapter 5 — Duplicates and inconsistency

Duplicate rows and inconsistent encodings silently distort counts, joins, and models

Learning objectives

- Explain duplicate impact on aggregates and training
- Describe how free-text categories hide structure

Duplicate data — causes and impact

- Duplicates arise from system retries, merged exports, copy-paste ETL
- Example 5.14 shows duplicate purchase rows
- Deduplication is not cosmetic, it changes business numbers

Example 5.14 — Duplicate customer purchase rows

- Example 5.14 — hands-on module
- The retail retry pattern
- Explore the chapter example module
- View files: ``modules/chapter5/example14/``

Inconsistent data — formats and units

- Inconsistent data is common when sources span departments or time periods
- Example 5.18 mixes date formats, gender encodings, and measurement units in one table
- Example 5.19 shows customer ID formats that block joins
- Example 5.20 splits Male versus M into separate categories
- Example 5.21 shows weeks lost normalizing free-text cities

Example 5.18 — Mixed dates, gender, and units

- Example 5.18 — hands-on module
- A checklist of silent parse failures
- Explore the chapter example module
- View files: ``modules/chapter5/example18/``

Join and aggregation failures

- Inconsistent keys prevent reliable joins across tables
- Mixed currencies or units make sums meaningless until converted
- Split labels inflate cardinality and starve rare categories of support
- Exploration in Chapter 6 surfaces many of these patterns

Takeaways

- Duplicates change revenue, sample weights, and runtime
- Inconsistency breaks time logic and joins
- Standardize formats early with explicit parsing rules and canonical category maps

Next

- Complete the quiz for this part
- Continue to outliers, irrelevant features, and class imbalance